TFRA-YOLO: Textile Feature Refinement and Reassembly YOLO for Closed-Loop Fabric Defect Inspection on Circular Knitting Machines

Kangzhao Tan, Jiarui Zhang, Junjie Yu, Heng Wu, and Chunhua He

***Abstract*—Circular knitting machines are foundational production infrastructure in the textile industry. A fabric defect arising during production can propagate during continuous knitting and cause substantial fabric loss if it is not detected and handled promptly. To address this issue, we design and implement a cloud-edge collaborative inspection system integrating a Hall sensor, adjustable supplementary lighting, an industrial high-definition camera, an RK3566 edge terminal, a textile-specific detector, and an IoT cloud platform. When the Hall sensor indicates that the machine is operating, the edge terminal activates the camera and supplementary light. Once a defect is detected, it stops the machine through the local wired control path and reports the defect to the cloud platform for defect statistics and production-quality analysis. The proposed TFRA-YOLO starts from YOLOv8n and combines directional texture encoding (DTE) with local texture-guided feature reassembly (LTFR) to retain directional knitted-texture cues and adaptively reconstruct the feature pyramid. On images collected from a real textile production line, TFRA-YOLO achieves 66.3% AP and 96.9% AP50, improving YOLOv8n by 7.0 and 2.8 percentage points, respectively. The proposed system provides a practical route from delayed post-production inspection toward traceable, in-process defect containment for circular knitting.**

***Index Terms*—Industrial defect detection, YOLO, deep learning, edge computing, Internet of Things.**

I. Introduction

CIRCULAR knitting machines are foundational production infrastructure in the textile industry: they continuously manufacture the knitted fabrics used in T-shirts, sportswear, and other daily garments. The latest International Textile Manufacturers Federation (ITMF) statistics report 28,000 global shipments of large circular knitting machines in 2024 [1]. Conventional inspection is commonly performed after production. Consequently, a defect arising early in production may continue propagating as the machine operates, leading to substantial fabric waste before it is discovered. An effective inspection system must therefore detect and localize defects during machine operation with sufficiently low latency to enable timely intervention before defective fabric continues to accumulate.

Single-stage YOLO detectors are particularly attractive for this setting because they jointly provide real-time inference and explicit defect localization. YOLO-based detectors have been adopted for industrial surface inspection of steel strips, steel surfaces, and large stamped parts [2]–[4]. In fabric-defect detection, existing studies have adapted YOLO through lightweight or tiny-model designs [5]–[7], adaptive or deformable feature extraction [8], [9], and enhanced multi-scale fusion or prediction for weak and small defects [10]–[16]. These studies establish the suitability of the YOLO route for bounding-box localization under industrial latency and resource constraints. However, most existing modifications primarily strengthen generic feature representations rather than explicitly exploiting the directional texture structure of circular-knitted fabrics.

For circular-knitted fabric inspection, two texture-specific limitations remain in conventional YOLO feature encoding and top-down reconstruction. First, circular-knitted fabrics exhibit highly repetitive texture patterns with dominant horizontal and vertical orientations. Defects typically disrupt this regular texture in horizontally extended, vertically extended, or compact local forms such as holes. Standard convolutional feature extraction can learn these patterns implicitly, but it does not explicitly introduce an orientation-aware inductive bias for separately modeling directional and compact texture disruptions. Second, nearest-neighbor upsampling in the top-down feature pyramid reconstructs higher-resolution features with a content-independent copying rule rather than adapting reconstruction weights to local texture structure. Such content-independent reconstruction may inadequately preserve narrow defect boundaries or weak local disruptions embedded in repetitive texture. These observations motivate explicit directional texture encoding during feature extraction and content-adaptive reconstruction during top-down fusion.

Based on this motivation, we develop TFRA-YOLO to introduce texture-specific inductive biases at both feature encoding and feature reconstruction stages. To address the first limitation, we introduce a directional texture encoding (DTE) module after selected backbone stages. DTE separately models horizontal, vertical, and compact local texture evidence, while residual fusion retains the baseline representation and augments it with directional texture cues. To address the second limitation, we introduce a local texture-guided feature reassembly (LTFR) pyramid that replaces fixed nearest-neighbor top-down upsampling with location-specific feature reassembly. TFRA-YOLO preserves the lightweight single-stage YOLO detection paradigm while introducing directional texture encoding in the backbone and content-adaptive reconstruction in the top-down feature pyramid.

Beyond detector design, practical deployment on circular knitting machines requires inspection to be synchronized with machine rotation and to support low-latency local intervention. We therefore integrate TFRA-YOLO into a cloud-edge in-

K. Tan, J. Zhang, J. Yu, and C. He are with the School of Computer Science, Guangdong University of Technology, Guangzhou 510006, China. H. Wu is with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China. Corresponding author: C. He (e-mail: hechunhua@pku.edu.cn). E-mails: K. Tan (tankangzhao@mails.gdut.edu.cn), J. Zhang (zhangjiarui1@mail2.gdut.edu.cn), J. Yu (yujunjie2@mails.gdut.edu.cn), H. Wu (heng.wu@foxmail.com), and C. He (hechunhua@pku.edu.cn).

spection system that combines image acquisition synchronized with machine rotation, RK3566-based edge inference, and cloud-based IoT monitoring. The latency-critical intervention loop remains entirely local, while defect records and device states are synchronized with the cloud IoT platform for status visualization and production-quality analysis.

The main contributions of our work are as follows.

- We introduce a directional texture encoding (DTE) module that explicitly models horizontal, vertical, and compact local texture cues and fuses them with the baseline backbone representation, providing a lightweight texture-specific inductive bias for circular-knitted defect detection.
- We develop a local texture-guided feature reassembly (LTFR) pyramid that replaces fixed nearest-neighbor upsampling with location-adaptive reconstruction, enabling content-adaptive reconstruction of local defect structures during top-down multi-scale fusion.
- We implement a cloud-edge inspection prototype that keeps the latency-critical intervention loop local while supporting centralized defect statistics and device supervision.

II. Method

A. TFRA-YOLO Architecture

TFRA-YOLO is built on YOLOv8n [17] and consists of three main components: a directional texture encoding backbone, a local texture-guided reassembly pyramid, and a multi-scale detection head. YOLOv8n is selected as the baseline because it provides favorable inference efficiency on the target RK3566 platform while retaining sufficient architectural flexibility for texture-specific modification. [18] As shown in Fig. 1, the directional texture encoding backbone introduces DTE after selected C2f stages to strengthen the representation of horizontally extended, vertically extended, and compact local texture disruptions. The local texture-guided reassembly pyramid replaces the fixed nearest-neighbor upsampling operations with LTFR, allowing higher-resolution features to be reconstructed according to local content before multi-scale fusion. The fused pyramid features are then forwarded to the detection head for defect classification and localization. The two proposed components target different stages of feature processing. DTE focuses on extracting direction-sensitive texture cues from the repetitive knitted background, whereas LTFR focuses on reconstructing and propagating these cues across feature scales.

B. Directional Texture Encoding

The visual cue of a defect in circular-knitted fabric is often a local disruption of repetitive texture rather than an isolated object boundary. Such disruptions commonly appear in horizontally elongated, vertically elongated, or compact local forms. A standard C2f block can learn these patterns implicitly through local feature aggregation, but it does not explicitly separate directional and compact texture responses. DTE is therefore introduced to encode these complementary structures with horizontal, vertical, and local branches. For a C2f output $X \in \mathbb{R}^{C\times H\times W}$, DTE first forms three depthwise branch responses:

$$F_h = \mathrm{DW}_{1\times 7}(X), \tag{1}$$

$$F_v = \mathrm{DW}_{7\times 1}(X), \tag{2}$$

$$F_l = \mathrm{DW}_{3\times 3}(X). \tag{3}$$

The 1×7 and 7×1 branches enlarge the receptive field along the horizontal and vertical directions, respectively, while the 3×3 branch preserves compact local texture information. The three responses are concatenated and fused by a 1×1 convolution:

$$F_d = \phi\left([F_h, F_v, F_l]\right), \tag{4}$$

where $[\cdot]$ denotes channel concatenation and ϕ denotes pointwise convolution followed by normalization and activation. Depthwise filtering keeps the directional branches lightweight, while the pointwise operation mixes the branch responses across channels. The fused response is added to the original C2f feature through a residual connection:

$$Y = X + F_d. \tag{5}$$

This preserves the baseline representation while introducing additional direction-sensitive texture cues with limited computational overhead. Although DTE is related to asymmetric convolution [19], it differs in using depthwise horizontal, vertical, and local branches together with three-way fusion and residual integration within the C2f backbone.

C. Local Texture-Guided Feature Reassembly

In the feature pyramid, fixed nearest-neighbor upsampling reconstructs higher-resolution features using a content-independent copying rule, without adapting to local texture structure. This can be suboptimal for multi-scale fusion in repetitive knitted textures, where narrow defect boundaries may occupy only a small portion of the local region. LTFR therefore replaces the two fixed upsampling operations with content-adaptive feature reassembly.

LTFR adopts content-aware feature reassembly to reconstruct each output location from a locally predicted neighborhood. This formulation follows the content-aware reassembly principle introduced in CARAFE [19]. Given $X \in \mathbb{R}^{C\times H\times W}$, a 1×1 compressor first reduces the kernel-prediction feature to $C_r = \max(C/4, 16)$ channels:

$$X_c = E_{1\times 1}(X). \tag{6}$$

A 3×3 encoder then predicts s^2k^2 kernel logits, where the scale factor is $s = 2$ and the reassembly kernel size is $k = 5$:

$$K' = E_{3\times 3}(X_c). \tag{7}$$

Pixel shuffle expands the logits to the upsampled grid,

$$\widehat{K} = \mathrm{PS}(K'), \tag{8}$$

and a softmax across the k^2 kernel positions produces a distinct normalized reassembly kernel at each output location:

$$K_{i,j} = \mathrm{softmax}(\widehat{K}_{i,j}). \tag{9}$$

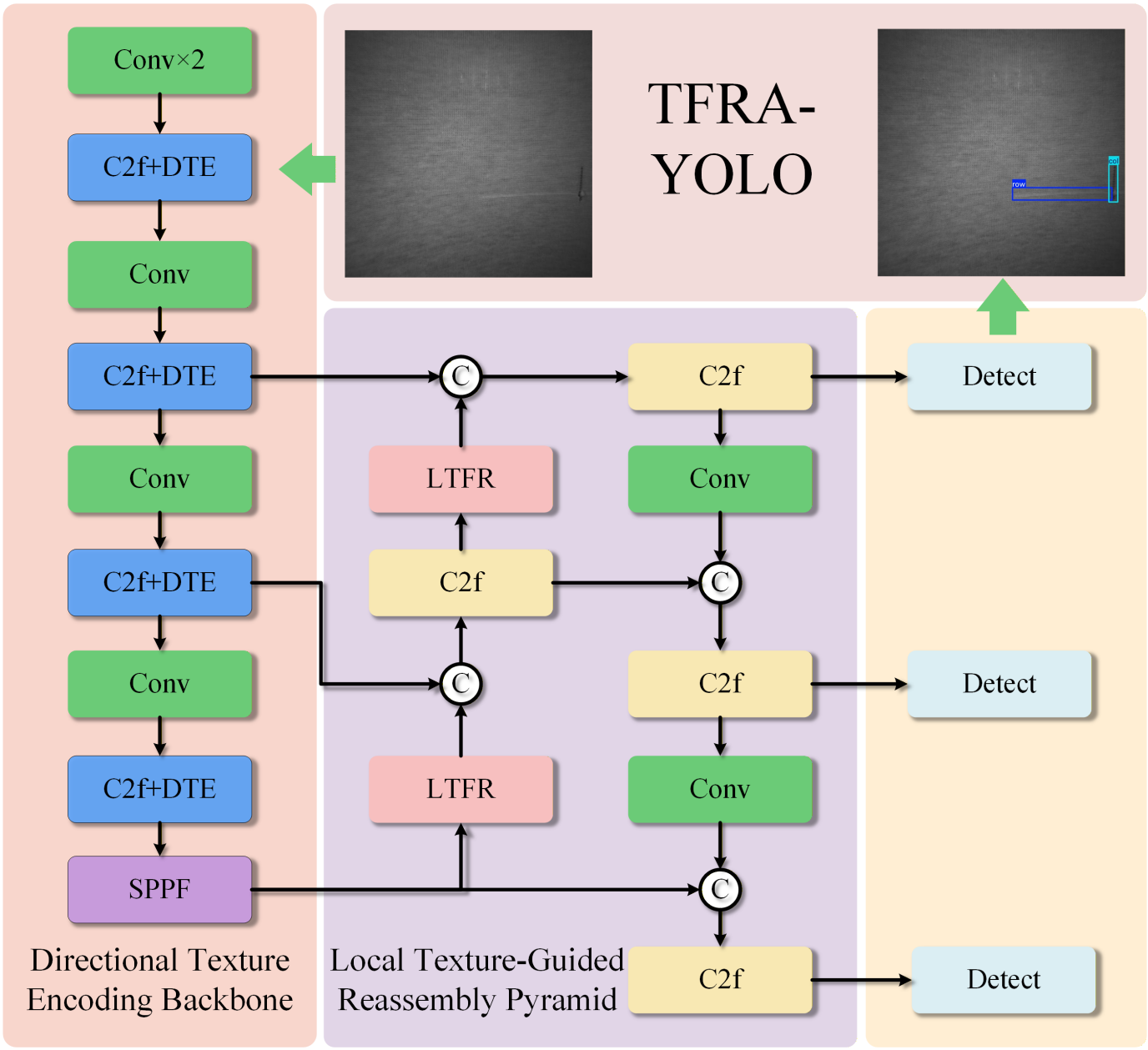

Fig. 1. Architecture of TFRA-YOLO. The Knitted Texture Fusion Backbone uses C2f–DTE blocks to encode horizontal, vertical, and local texture cues. The Local Texture Reassembly Pyramid replaces the two top-down fixed upsampling operations with LTFR, while the bottom-up path and three-scale detection head follow YOLOv8n.

The normalization constraint is

$$\sum_{u=1}^{k^2} K_{i,j,u} = 1. \tag{10}$$

Here, $E_{1\times1}$ and $E_{3\times3}$ denote the compression and kernel-encoding layers, respectively, and PS denotes pixel shuffle. In parallel, a local $k \times k$ feature patch is extracted from X and aligned with the enlarged grid:

$$P_{i,j} = \mathrm{Unfold}_{k\times k}(X)_{i,j}. \tag{11}$$

where $P_{i,j} \in \mathbb{R}^{C\times k^2}$ denotes the local feature neighborhood associated with output location (i, j). The reassembled feature is then obtained by the location-specific weighted sum

$$Y_{i,j} = \sum_{u=1}^{k^2} K_{i,j,u} P_{i,j,u}. \tag{12}$$

Unlike a spatially shared interpolation kernel, Eq. (12) allows the reconstruction weights to vary with local texture content, so defect boundaries and regular knitted regions can be aggregated differently. The channel-compression stage bounds the overhead of predicting these kernels, while the 5×5 neighborhood provides more local support than a copied source pixel. In TFRA-YOLO, this reassembly mechanism is applied specifically to the two feature-pyramid upsampling

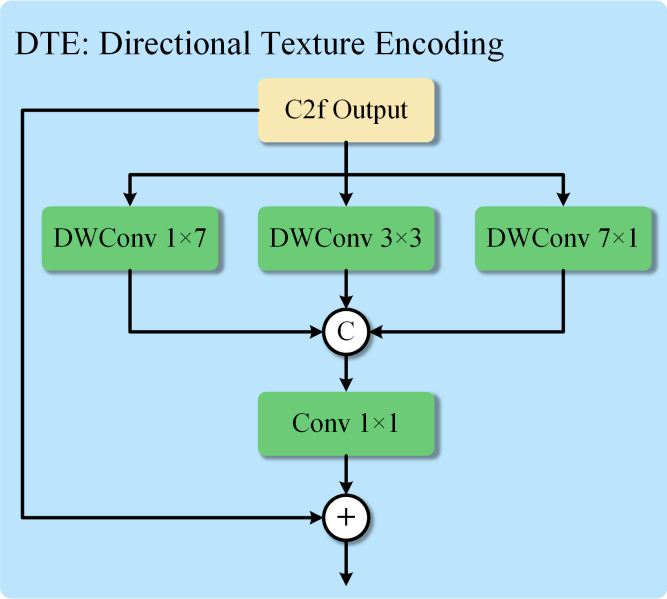

Fig. 2. DTE structure. A C2f feature is processed by horizontal, vertical, and local depthwise branches, followed by channel concatenation, 1×1 fusion, and residual addition.

stages to preserve local defect structure during multi-scale fusion.

DTE and LTFR play complementary roles: DTE enhances direction-sensitive texture representation in the backbone, while LTFR performs content-adaptive reconstruction of these features across pyramid scales.

D. Closed-Loop Inspection System

Fig. 4 shows the inspection system installed on a circular knitting machine, comprising a Hall sensor, an industrial camera, and an RK3566 edge device. The workflow begins when the operator starts the machine. The Hall sensor detects machine rotation and signals the RK3566, which activates the supplementary illumination, acquires camera frames, and performs real-time defect inspection. Once a defect is detected, the RK3566 immediately issues a stop command through the wired connection to the machine controller. The corresponding defect frame is then displayed locally for operator inspection and troubleshooting. After the problem is resolved and the machine is restarted, the inspection cycle resumes. The complete defect-to-stop loop is executed locally and does not depend on cloud communication.

To support real-time inference on the RK3566 NPU, the trained detector is converted from the training framework to an edge-executable representation. The PyTorch model is first exported to ONNX at an input resolution of 640×640, and then converted to RKNN using the official RKNN-Toolkit2 [20]. Post-training INT8 quantization is applied during conversion, with representative training images used for calibration [21]. The resulting RKNN model is loaded by the Android edge application and executed on the RK3566 NPU. The deployment pipeline is therefore

$$\text{PyTorch} \rightarrow \text{ONNX} \rightarrow \text{INT8 RKNN} \rightarrow \text{RK3566 NPU}.$$

TABLE I
IMAGE-LEVEL SPLIT AND DEFECT-INSTANCE COUNTS IN THE REAL PRODUCTION-LINE DATASET.

Split	Images	Instances	Row	Col	Hole
Train	282	329	174	75	80
Validation	40	48	24	13	11
Test	81	91	60	12	19

While defect detection and stop control are handled locally, non-latency-critical information is synchronized to the cloud platform. As shown in Fig. 5, the cloud IoT platform receives defect records and device states from the edge terminal for centralized visualization, defect statistics, and production-quality analysis.

III. Experimental Setup

A. Dataset

The dataset was collected from a real circular-knitting production line using the deployed image-acquisition system described in Section II-D. Images were captured during machine operation and manually annotated with axis-aligned bounding boxes by the project team. The dataset contains three engineering defect categories: row, col, and hole, which generally appear as horizontal disruptions, vertical disruptions, and compact local damage, respectively. A total of 403 images containing 468 defect instances were divided into training, validation, and test sets with 282, 40, and 81 images, respectively. The corresponding instance statistics are listed in Table I. The validation set was used for checkpoint selection, whereas all quantitative results in this paper were obtained on the held-out test set containing 81 images and 91 instances. The implementation protocol and training hyperparameters are summarized in Table II.

B. Evaluation Metrics

We use the COCO detection protocol [20]. For a predicted box B_p and a ground-truth box B_g, intersection over union (IoU) is defined as $\mathrm{IoU}(B_p, B_g) = |B_p \cap B_g| / |B_p \cup B_g|$. At an IoU threshold t, a prediction is counted as a true positive only when it has the correct class and is matched one-to-one to an as-yet unmatched ground-truth instance; all remaining predictions are false positives and unmatched ground-truth instances are false negatives. For each class c, $\mathrm{AP}_c(t)$ is the area under its interpolated precision–recall curve. With $C = 3$ classes, the reported AP and AP50 are

$$\mathrm{AP} = \frac{1}{10C} \sum_{c=1}^{C} \sum_{t \in \mathcal{T}} \mathrm{AP}_c(t),$$
$$\mathrm{AP50} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{AP}_c(0.50), \tag{13}$$

where $\mathcal{T} = \{0.50, 0.55, \ldots, 0.95\}$. Thus, AP measures localization and classification quality across increasingly strict overlap thresholds, whereas AP50 emphasizes detection at $\mathrm{IoU} = 0.50$. Both are written as percentages in the tables.

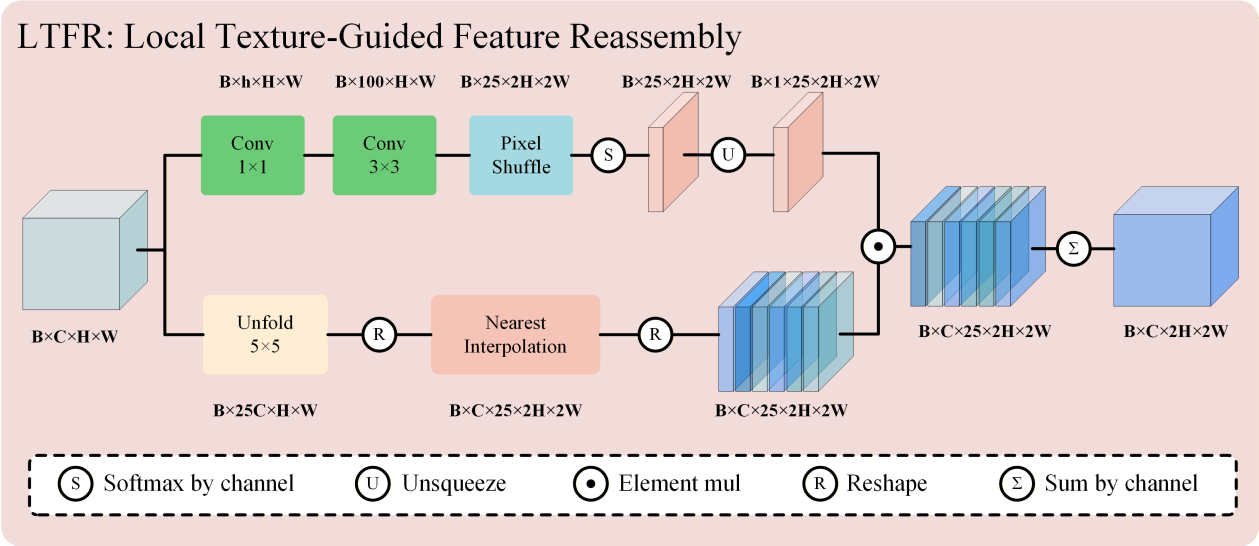

Fig. 3. LTFR structure. The upper path predicts location-specific 5×5 reassembly kernels, and the lower path unfolds and aligns local feature patches. Their element-wise product is summed along the kernel dimension to produce a $2\times$ upsampled feature.

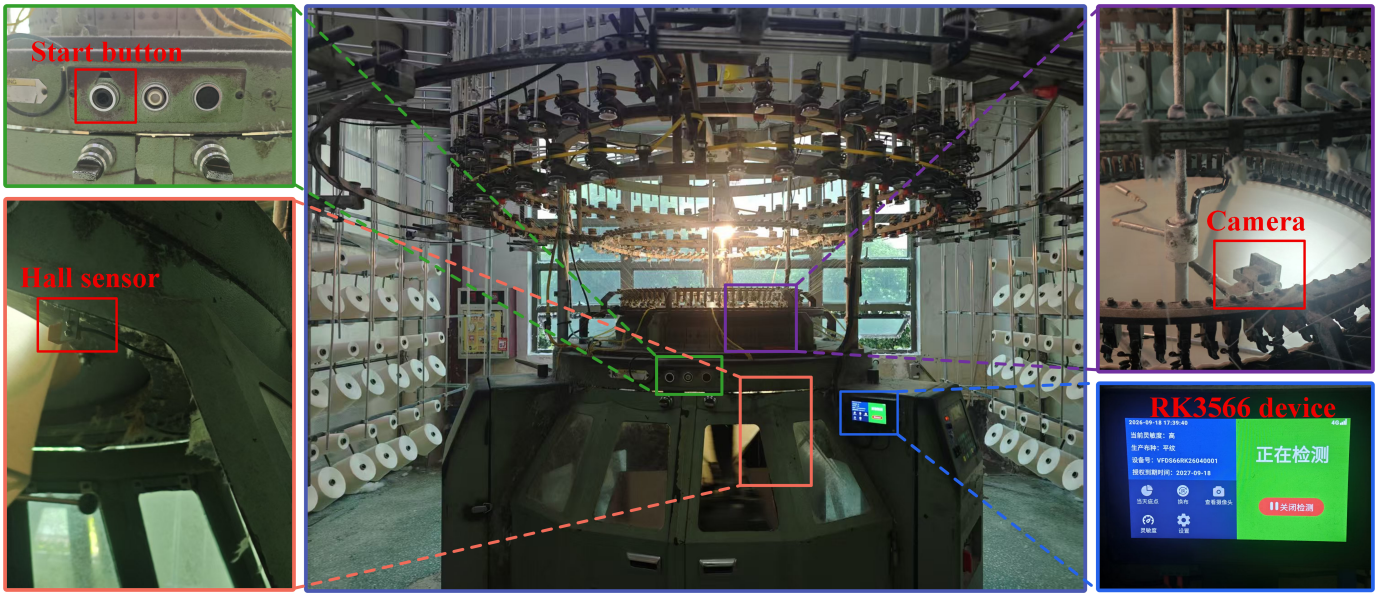

Fig. 4. Actual deployment on the circular knitting machine. The insets highlight the start button, Hall sensor, industrial camera, and RK3566 edge device, respectively.

We additionally report trainable parameters and FLOPs at $[1, 3, 640, 640]$ to characterize model scale and computational cost; these are not substitutes for measured edge-device latency.

For each model, the retained best-validation checkpoint is evaluated once on the same held-out test split. FLOPs and parameter counts are reported at $[1, 3, 640, 640]$. The values are suitable for a coarse efficiency comparison, but cross-framework FLOP accounting can omit unsupported operators; no cross-framework latency claim is made. DETR, Deformable DETR, Faster R-CNN, and RT-DETRv2 follow their respective repositories and pretrained backbones [21]–[24].

IV. Results and Discussion

A. Comparison With Detectors

Table III compares TFRA-YOLO with the retained baselines. TFRA-YOLO achieves 66.3% AP and 96.9% AP50. Relative to its direct YOLOv8n baseline, AP improves by 7.0 points and AP50 by 2.8 points for 0.38 M additional parameters and 0.8 G additional FLOPs. Among the compact YOLO variants evaluated here, TFRA-YOLO has the highest AP. RT-DETR-R18 obtains 67.7% AP, 1.4 points higher than TFRA-YOLO, but requires 20.09 M parameters and 60.72 G FLOPs. This is a useful trade-off rather than a claim that TFRA-YOLO dominates every detector.

Fig. 6 visualizes the parameter- and operation-accuracy trade-off without shifting any data coordinates. The position

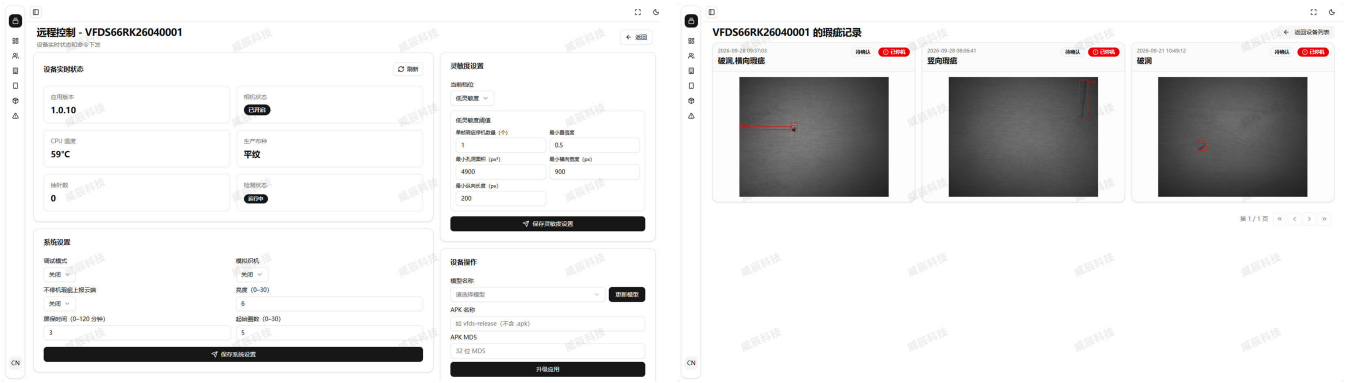

Fig. 5. Cloud IoT platform for the deployed inspection system. The four panels show the platform portal, defect-count trend dashboard, device-state and remote-configuration interface, and uploaded defect-record review interface, respectively. Cloud communication supports status visualization, statistics, and authorized supervision, whereas the defect-to-stop control path remains local and wired.

TABLE II
IMPLEMENTATION PROTOCOL AND TRAINING HYPERPARAMETERS.

Item	Specification
Training hardware	NVIDIA GeForce RTX 3090 GPU (24 GB)
Software	Ubuntu 24.04, Python 3.10.20, PyTorch 2.4.1+cu124, and Ultralytics 8.4.140
Model initialization	TFRA-YOLO and YOLOv8n: `yolov8n.pt`; other baselines: the available official pretrained checkpoint in the corresponding repository
Input resolution	640×640
Batch size / data workers	16 / 8
Maximum epochs / early stopping patience	300 / 80 epochs
Optimizer	Ultralytics automatic optimizer selection
Data augmentation	Standard Ultralytics geometric and photometric augmentations; no augmentation hyperparameter was manually overridden
Checkpoint retained for testing	Framework-selected best validation checkpoint
Test protocol	Held-out test split, maximum 16 detections per image

TABLE III
TEST-SET COMPARISON ON 81 IMAGES AND 91 INSTANCES. AP IS COCO AP@[.50:.95]. ALL VALUES ARE PERCENTAGES EXCEPT PARAMETERS AND FLOPS. BOLD AND UNDERLINED VALUES DENOTE THE BEST AND SECOND-BEST AP/AP50 RESULTS, RESPECTIVELY.

Model	Params (M)	FLOPs (G)	AP50 (%)	AP (%)
Lightweight Real-Time Detectors				
YOLOv5n	2.50	7.10	93.1	58.4
YOLO11n	2.58	6.40	95.0	59.6
YOLO12n	2.56	7.30	94.3	60.9
YOLO26n	2.38	5.30	96.5	61.4
End-to-End Detectors				
RT-DETR-R18	20.09	60.72	95.2	**67.7**
DETR-R50	41.28	77.25	96.4	53.6
Deformable DETR-R50	39.82	158.66	<u>96.8</u>	64.3
Two-Stage Detector				
Faster R-CNN-R50	41.13	181.45	96.1	64.7
Baseline				
YOLOv8n	3.01	8.10	94.1	59.3
Ours				
TFRA-YOLO	3.39	8.90	**96.9**	<u>66.3</u>

of TFRA-YOLO in the low-parameter region is consistent with the tabulated result. Because the horizontal axes are logarithmic, the visual is intended to show scale rather than imply a linear performance-efficiency relationship.

B. Ablation

Table IV isolates the two modifications within the same YOLOv8n family. DTE contributes the larger isolated improvement (+4.8 AP), while LTFR contributes +1.6 AP. Com-

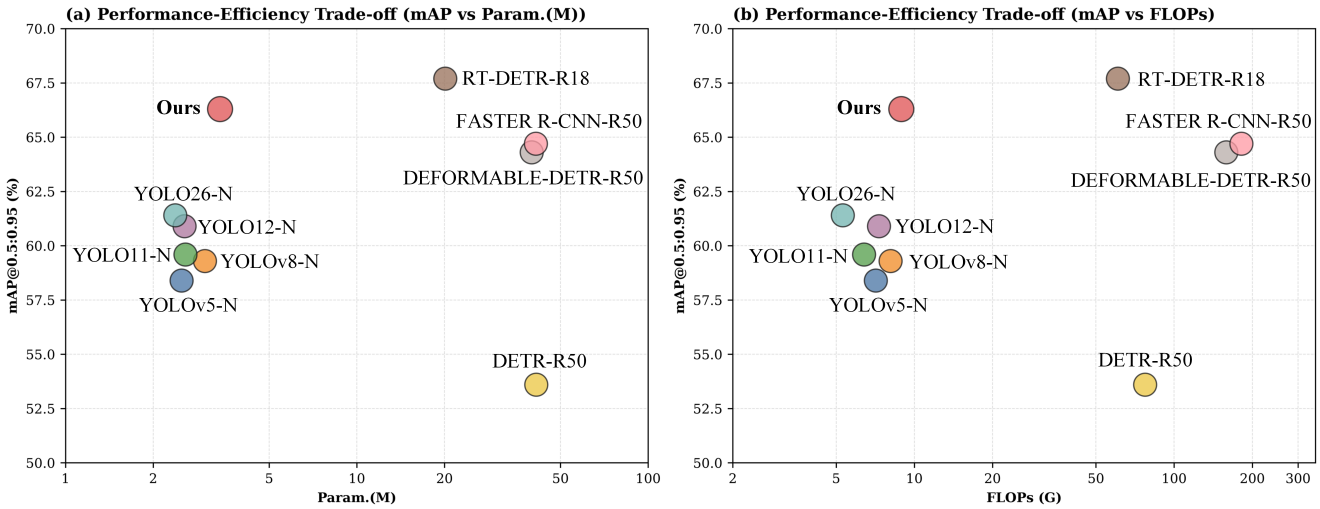

Fig. 6. Accuracy-efficiency trade-off of the retained test results. Both horizontal axes use a logarithmic scale. Marker coordinates are the exact values in Table III; the plot does not contain manual coordinate compression.

TABLE IV
ABLATION ON THE COMMON HELD-OUT TEST SPLIT USING THE SAME YOLOV8N TRAINING RECIPE.

Variant	DTE	LTFR	Params	FLOPs	AP50	AP
YOLOv8n	–	–	3.01	8.1	94.1	59.3
+ DTE	✓	–	3.28	8.8	94.2	64.1
+ LTFR	–	✓	3.11	8.2	94.3	60.9
TFRA-YOLO	✓	✓	3.39	8.9	96.9	66.3

bining both gives 66.3% AP, which is 0.6 points above the arithmetic sum of the isolated gains. This indicates that the texture representation and adaptive reassembly are complementary: DTE supplies direction-specific texture cues, whereas LTFR transfers them through the top-down path. The current study establishes module-level complementarity; branch-level DTE and alternative-upsampling ablations are required before making more granular claims about individual directional branches, the directional kernel length, or the choice of content-adaptive reassembly.

C. Per-Class and Qualitative Results

The TFRA-YOLO checkpoint obtains 61.7%, 73.8%, and 63.4% AP for *row* (horizontal), *col* (vertical), and *hole* defects, respectively. The notably high *col*-defect AP should be interpreted with care because the test category has only 12 instances. Fig. 7 presents seven selected images and the available standardized renderings. The visual comparison is illustrative rather than a replacement for the full quantitative test set. It is particularly useful for examining thin *col*-type defects, long *row*-type defects, and holes against repetitive background texture.

Fig. 8 further compares the activation visualizations produced by the official Ultralytics visualization routine for the YOLOv8n baseline and TFRA-YOLO. Each row corresponds to the same test image, with the original image, baseline visualization, and TFRA-YOLO visualization arranged from left to right. These maps are used only as qualitative evidence of where the detectors respond; the AP results in Tables III and IV remain the basis for the quantitative comparison.

D. Limitations and Submission-Ready Validation

The present evidence supports a promising lightweight detector on a real production-line image set, but it does not yet establish production reliability. First, the test split is small and category-imbalanced, and its image-level partition does not establish isolation of adjacent video frames. Second, FLOPs are not a substitute for end-to-end edge latency, especially when image capture, preprocessing, postprocessing, control communication, and physical stopping are part of the operational loop. Third, the current ablation isolates the two modules but does not yet separately quantify each DTE branch or compare LTFR with alternative interpolation and reassembly operators. Finally, operator troubleshooting records can support defect statistics and production-quality analysis, but their completeness and operational value have not yet been measured. These points define a concrete validation plan, rather than weaknesses that should be concealed.

V. CONCLUSION

This work presents a closed-loop framework for defect inspection on a circular knitting machine and TFRA-YOLO, a compact direction- and content-adaptive detector. The workflow couples rotation-aware edge inspection, automatic local wired stop signaling, on-site operator inspection and troubleshooting, and a cloud digital twin for live status visualization, defect statistics, production-quality analysis, and authorized remote supervisory control. The detector combines DTE for horizontal, vertical, and local knitted-texture encoding with LTFR for adaptive feature-pyramid reconstruction. On the retained test split, TFRA-YOLO improves YOLOv8n from 59.3% to 66.3% AP at 3.39 M parameters and 8.9 G FLOPs. The next step is not merely more offline accuracy: it is a

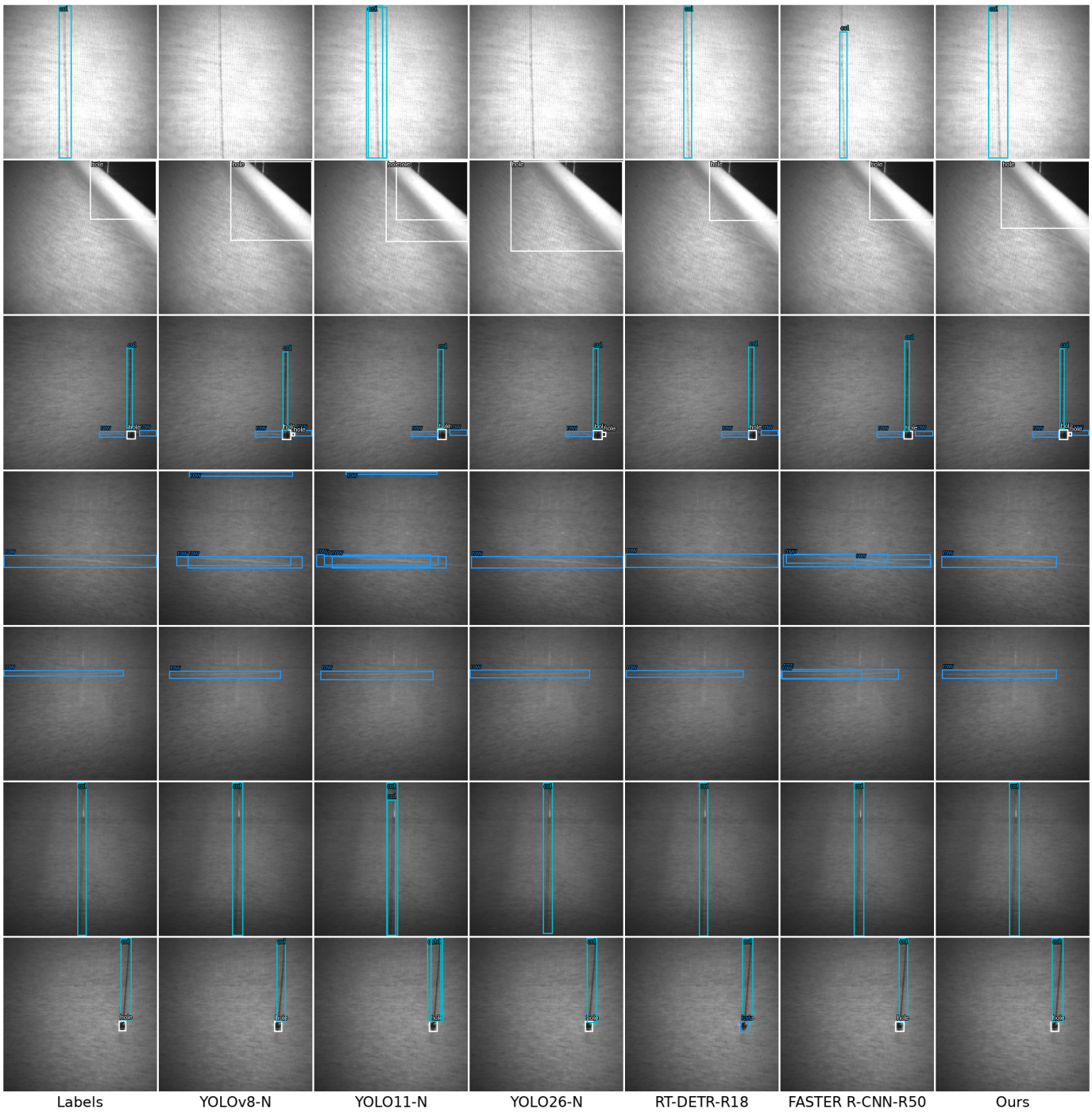

Fig. 7. Qualitative comparison on selected test images. Columns are labels, YOLOv8n, YOLO11n, YOLO26n, RT-DETR-R18, Faster R-CNN-R50, and TFRA-YOLO. The figure uses a common renderer for visual consistency; it should be updated after finalizing the exact sample-selection protocol and display confidence threshold.

disciplined site evaluation of sensing-to-stop latency, false-stop behavior, operator-troubleshooting workflow, production-quality-analysis value, and grouped generalization across production conditions.

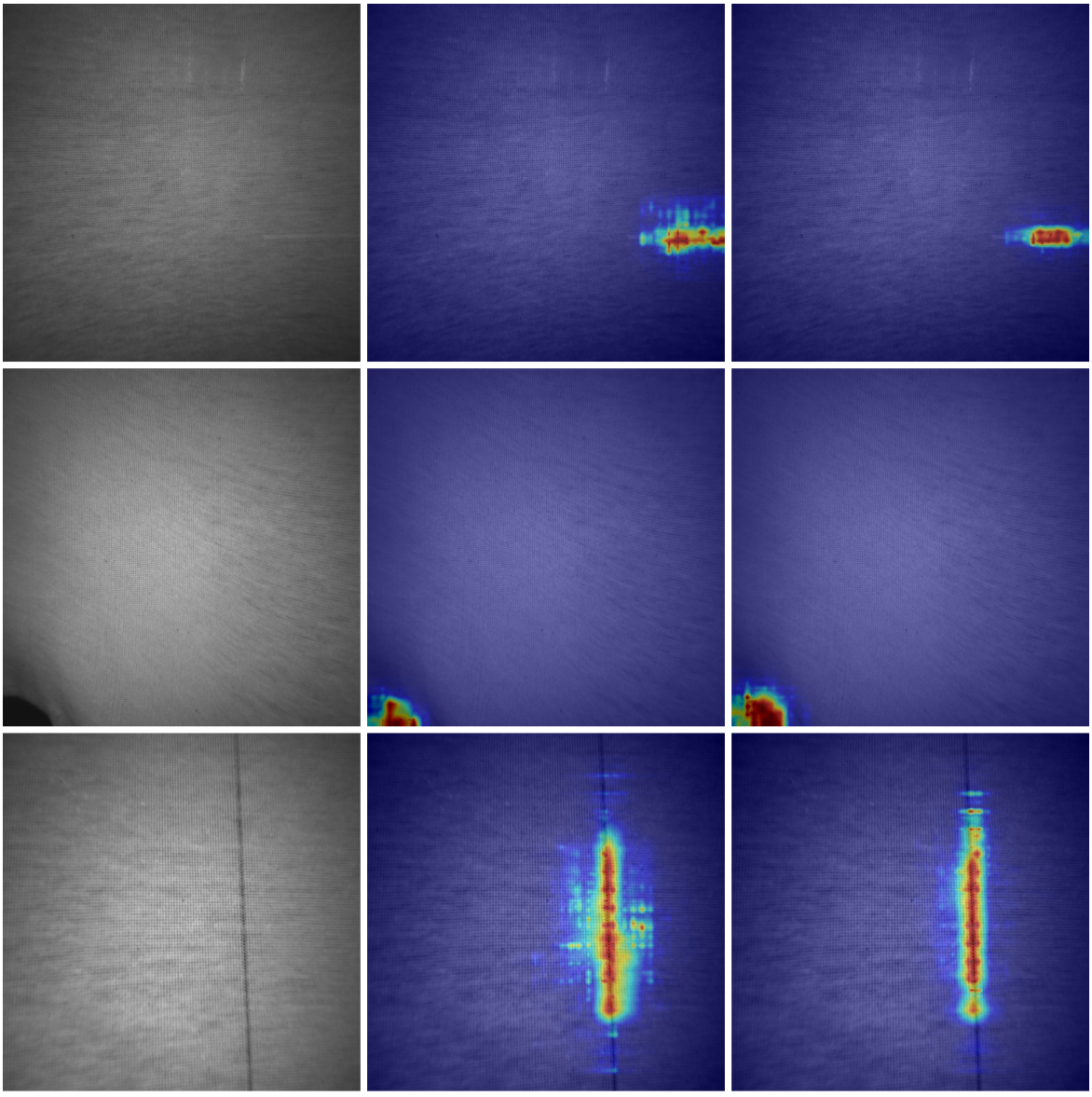

Fig. 8. Qualitative activation visualization on three selected test images. The left, middle, and right columns show the original images, YOLOv8n visualizations, and TFRA-YOLO visualizations, respectively. Each visualization is generated with the same official Ultralytics routine.

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